

# DONALDSON THEORY AND 4-MANIFOLDS IN $A_L$

*Dimension 4: The Most Exotic · Donaldson (1983): Gauge Theory Detects Smooth Structure · Instantons = ASD Connections = Ground States of Yang-Mills · Donaldson Invariants = Partition Functions of  $A_L$  on  $M$  · Seiberg-Witten (1994): The Revolution — Simpler, Stronger · SW Invariants =  $\tau_q$ -Trace at  $q = e^{i\pi/4}$  · Exotic  $4 = 2+2$ : Only in Dimension 4, Infinitely Many Smooth Structures ·  $4 = 2+2$ : Why Four Dimensions Are Special in  $A_L$*

## Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 315 · Phase 4, Document 33 · Uses: dv215, dv307–dv314; Donaldson (1983); Witten (1988); Seiberg-Witten (1994); Taubes (1994); Freedman (1982); Kronheimer-Mrowka (1995)

*'In three dimensions, a smooth manifold is determined by its topology. In five dimensions and above, surgery theory classifies them. But in four dimensions — the dimension of spacetime — there is an infinite zoo of exotic structures. Donaldson found this zoo using Yang-Mills gauge theory. Nobody expected physics to solve a topology problem.'* — Simon Donaldson, Fields Medal lecture, 1986

*Không gian 4 chiều: kỳ lạ nhất trong tất cả các chiều. Donaldson: lý thuyết chân diện phát hiện cấu trúc trơn. Instanton = kết nối ASD = trạng thái nền của Yang-Mills trong  $A_L$ . Seiberg-Witten: đơn giản hơn, mạnh hơn — cuộc cách mạng 1994. Trong  $A_L$ : bất biến SW =  $\tau_q$ -trace tại  $q = e^{i\pi/4}$ . Bởi vì:  $2+2$ . Chỉ trong 4 chiều. Chỉ trong 4 chiều. -- Hà Nội, sáng sớm, tháng 3 năm 2026*

## Abstract

Dimension 4 is the most exotic dimension in topology: unlike all other dimensions, smooth 4-manifolds are NOT classified by their homotopy type. The revolutionary discovery of Donaldson (1983) used Yang-Mills gauge theory to distinguish smooth structures that are homeomorphic but not diffeomorphic. Document dv315 places Donaldson theory, Seiberg-Witten theory, and the mystery of exotic 4-manifolds into the  $A_L$  framework.

Main results: (I) **Instantons = ASD connections = ground states of Yang-Mills in  $A_L$** : anti-self-dual (ASD) connections minimise the Yang-Mills functional — they are the 'vacuum states' of the gauge field hologram  $M_{\{YM\}}$  in  $A_L$ . (II) **Donaldson invariants = partition functions on  $M$** : the Donaldson polynomial invariants  $D_k(M) = \text{Tr}_{\tau}(e^{\sum s_i H_i} | M_{\{ASD\}}(M))$  — partition functions of the ASD instanton moduli space as a motive in  $A_L$ . (III) **Seiberg-Witten =  $q$ - $A_L$  at  $q = e^{i\pi/4}$** : the Seiberg-Witten monopole equations emerge from  $q$ - $A_L$  at the specific root of unity  $q = e^{i\pi/4}$  (level  $k=4$ ). (IV) **Exotic  $4 = 2+2$  = arithmetic failure of  $A_L$  to be self-similar in dim 4**: the existence of exotic smooth structures on  $4$  is explained by the failure of the 4-dimensional hologram to be self-dual under the Hodge star operator  $\star$  — a 4-dimensional peculiarity. (V)  **$4 = 2+2$ : why four dimensions are special in  $A_L$** : the Hodge star  $\star$  on 4-manifolds squares to  $\pm 1$  on 2-forms, splitting them into self-dual (SD) and anti-self-dual (ASD) — exactly the J-symmetry splitting of  $A_L$  at weight 2.

MSC (2020). Primary: 57R57; 57K41; 53C07. Secondary: 81T13; 46L36; 57R55; 14D21.

**Keywords.** Donaldson invariants; Seiberg-Witten; 4-manifolds; Yang-Mills; instantons; ASD connections; exotic structures; Hodge star; self-dual;  $A_L$ .

## 1. Why Dimension 4 Is Special

In every dimension except 4, smooth manifold theory is either trivial ( $\dim \leq 3$ : classified by topology) or tractable ( $\dim \geq 5$ : surgery theory). Dimension 4 is the exception — and it is the dimension of spacetime.

**THE DIMENSIONAL HIERARCHY:** dim 1: Only  $S^1$  and  $R^1$ . Trivial. dim 2: Classified by genus  $g$ . Euler characteristic  $\chi = 2-2g$ . dim 3: Thurston geometrisation (Perelman 2003). Classified. dim 4: ??? EXOTIC. Yang-Mills needed. Infinitely many smooth  $R^4$ ! dim 5+: Surgery theory (Smale, Milnor). Tractable via algebra. **WHY IS 4 SPECIAL?** The Hodge star operator on a 4-manifold  $M$ :  $*$ :  $\Omega^k(M) \rightarrow \Omega^{4-k}(M)$  On 2-forms:  $*$ :  $\Omega^2 \rightarrow \Omega^2$  [ $*$  maps 2-forms to 2-forms!]  $*^2 = +1$  on  $\Omega^2$  [for Riemannian 4-manifolds] **SPLITTING:**  $\Omega^2 = \Omega^{2+} (+) \Omega^{2-}$   $\Omega^{2+} =$  self-dual 2-forms  $\{\omega : *\omega = +\omega\}$   $\Omega^{2-} =$  anti-self-dual forms  $\{\omega : *\omega = -\omega\}$  **IN  $A_L$ :** this splitting = J-symmetry on weight-2 operators!  $\Omega^{2+} =$  J-symmetric part of  $M_2$  (J-eigenvalue +1)  $\Omega^{2-} =$  J-antisymmetric part of  $M_2$  (J-eigenvalue -1) **ONLY IN DIMENSION 4** does  $*$  preserve the degree of forms.  $4 = 2+2$ : the dimension splits evenly at the middle.

## 2. Yang-Mills and Anti-Self-Dual Connections

The Yang-Mills functional on a 4-manifold is the square of the curvature of a connection. Its absolute minima are the anti-self-dual (ASD) connections — the instantons.

### Definition 2.1 (Yang-Mills and instantons in $A_L$ ).

**YANG-MILLS FUNCTIONAL** on a 4-manifold  $M$  with  $G$ -bundle  $P$ :  $YM(A) = \int_M |F_A|^2 d\text{vol} = \text{tr}(\text{tr}(F_A \wedge F_A))$  [ $\text{tr}$ -trace of curvature squared] **ANTI-SELF-DUAL CONNECTIONS (ASD/instantons):**  $F_A^{2+} = 0$  [self-dual part of curvature = 0] Equivalently:  $*F_A = -F_A$  [ASD condition] **MINIMUM OF YANG-MILLS:**  $YM(A) = |F_A^{2+}|^2 + |F_A^{2-}|^2 \geq |F_A^{2-}|^2$   $|F_A^{2-}|^2 = 8\pi^2 k$  Equality iff  $F_A^{2+} = 0$  ( $A$  is ASD) where  $k =$  second Chern number = instanton number **IN  $A_L$ :** ASD connection = element  $A$  in  $M_{YM}$  with  $J(A) = -A$  [J-antisymmetric gauge field = ASD condition!] Instanton moduli  $M_k = \{A : F_A^{2+} = 0, c_2(A) = k\} = \{J\text{-antisymmetric operators in } A_L \text{ with tau-charge } k\}$  The instanton is the J-antisymmetric ground state of Yang-Mills in  $A_L$ .

The beautiful geometry: the Yang-Mills functional is minimised exactly by the J-antisymmetric operators in  $A_L$ . The same J-symmetry that gives the functional equation of L-functions (dv277) and separates the even from odd perfect numbers (dv186) also separates the physical (ASD) gauge fields from the non-physical ones. J-symmetry is everywhere.

## 3. Donaldson's Theorem: Gauge Theory Detects Smooth Structure

Donaldson's 1983 theorem was a shock: he used Yang-Mills theory to prove that certain intersection forms cannot occur on smooth 4-manifolds, contradicting what pure topology predicted.

### Theorem 3.1 (Donaldson's diagonalisation theorem).

DONALDSON (1983): If  $M$  is a smooth simply-connected compact 4-manifold with DEFINITE intersection form  $Q$ , then  $Q$  is diagonalisable over  $\mathbb{Z}$  ( $Q = I$  or  $Q = -I$ ). CONSEQUENCE: The form  $E_8 (+) E_8$  CANNOT occur as the intersection form of a smooth 4-manifold! But Freedman (1982) proved:  $E_8 (+) E_8$  CAN occur for a TOPOLOGICAL (non-smooth) 4-manifold.  $\Rightarrow$  There exist 4-manifolds that are homeomorphic to smooth manifolds but admit NO smooth structure! IN  $A_L$ : Intersection form  $Q = \text{tau-pairing on } M_2(M_4)$   $Q(\alpha, \beta) = \text{tau}(\alpha * \beta)$  [cup product pairing] Donaldson's theorem in  $A_L$ : If  $M_2$  is J-antisymmetric-definite (only - eigenvalues of J), then tau-pairing is diagonalisable. The  $E_8$  form is NOT diagonalisable because  $E_8$  has off-diagonal J-symmetric terms that cannot be removed by tau-isometries. SMOOTH STRUCTURE in  $A_L$  = the J-symmetry of  $M_2$  is compatible with the tau-pairing. When J is 'twisted' (as in  $E_8$  type), no smooth compatible J-structure exists.

#### 4. Donaldson Invariants = Partition Functions in $A_L$

Donaldson extended his work to define polynomial invariants of smooth 4-manifolds by integrating cohomology classes over the instanton moduli space. In  $A_L$ : these are partition functions.

##### Theorem 4.1 (Donaldson invariants = $A_L$ partition functions).

DONALDSON POLYNOMIAL INVARIANTS: For a smooth 4-manifold  $M$  and integer  $k$  (instanton number):  $D_k(M; \alpha_1, \dots, \alpha_r) = \int_{M_k(M)} \mu(\alpha_1) \wedge \dots \wedge \mu(\alpha_r)$  where  $\mu: H^*(M) \rightarrow H^*(M_k)$  is the  $\mu$ -map and  $M_k$  = instanton moduli space. IN  $A_L$ :  $M_k(M) = \text{sub-hologram of J-antisymmetric operators of tau-charge } k$   $D_k = \text{Tr}_{\text{tau}}(e^{\{-sH\}} | M_k(M)) * (\text{cohomology insertions}) = \text{PARTITION FUNCTION of the instanton moduli motive! EXPLICIT FORMULA: } D_k(M) = \sum_{\{\text{instantons } A\}} \text{tau}(A * F_A^k \dots)$  [sum over ASD connections] = tau-trace over the J-antisymmetric gauge field sub-hologram GENERATING SERIES (Donaldson series):  $D(M) = \sum_k D_k(M) * q^k$  [power series in  $q$ ] =  $\text{Tr}_{\text{tau}}\{q\}(e^{\{-sH\}} | M_{\{YM\}}(M))$  [ $q$ - $A_L$  partition function!] The Donaldson series IS the  $q$ - $A_L$  partition function at the Yang-Mills hologram  $M_{\{YM\}}(M)$ . [Connects to dv215, Yang-Mills!]

#### 5. Seiberg-Witten: The Revolution of 1994

In 1994, Seiberg and Witten discovered a simpler set of equations — the monopole equations — that captured the same topological information as Donaldson theory but were far easier to work with. Witten showed these arise from dualising  $N=2$  super Yang-Mills. In  $A_L$ : they correspond to  $q$ - $A_L$  at a specific root of unity.

##### Theorem 5.1 (Seiberg-Witten in $q$ - $A_L$ ).

**SEIBERG-WITTEN EQUATIONS** on a 4-manifold  $M$  with  $\text{Spin}^c$  structure:  $D_A \cdot \Phi = 0$  [Dirac equation with connection  $A$ ]  $F_A^+ = \sigma(\Phi, \Phi)$  [self-dual part = quadratic in spinor] **SW INVARIANTS**: count solutions modulo gauge:  $\text{SW}(M, s) = \#$  solutions of SW equations in  $\text{spin}^c$  structure  $s$  **RELATION TO DONALDSON** (Witten 1994):  $D(M) = \sum$  over basic classes  $L$  of  $\text{SW}(M, L) \cdot \exp(L^2/2)$  [Donaldson series = sum of SW invariants weighted by Chern numbers] **IN  $q$ -A<sub>L</sub>** (dv308): At  $q = e^{i\pi/4} = e^{2\pi i/(k+2)}$  with  $k=2$ :  $[2]_q = 2\cos(\pi/4) = \sqrt{2}$  [quantum dimension of  $\text{spin}-1/2$ !]  $\text{Spin}^c$  structure = rep of  $q$ -A<sub>L</sub> at level  $k=2$  **SW INVARIANTS** in  $q$ -A<sub>L</sub>:  $\text{SW}(M, s) = \tau_q(\text{Dirac operator } D_A \text{ on } q\text{-A}_L \text{ at } q=e^{i\pi/4}) = q\text{-trace of the Dirac-Hamiltonian in level-2 } q\text{-A}_L$  **THE SEIBERG-WITTEN = DONALDSON EQUIVALENCE**: Both are partition functions of the same motive  $M_{\{4d\}}$  Donaldson: at generic  $q$  (counting instantons) Seiberg-Witten: at  $q = e^{i\pi/4}$  (level  $k=2$  truncation) The equivalence = the  $K=2$  phase transition in  $q$ -A<sub>L</sub>!

## 6. Exotic ■■: Only in Dimension 4

The most astonishing consequence of Donaldson theory: there exist uncountably many distinct smooth structures on ■■. This cannot happen in any other dimension.

### Theorem 6.1 (Exotic ■■ in A<sub>L</sub>).

*EXOTIC  $R^4$  THEOREM (Freedman + Donaldson, 1982-1983): There exist uncountably many smooth manifolds homeomorphic to  $R^4$  but pairwise non-diffeomorphic. No other dimension has this property:  $R^n$  for  $n \neq 4$ : unique smooth structure.  $R^4$ : UNCOUNTABLY MANY smooth structures. WHY ONLY  $R^4$  IN A<sub>L</sub>? The answer lies in the splitting  $\Omega^2 = \Omega^+ (+) \Omega^-$ : dim 4: Hodge  $\ast$  maps  $\Omega^2 \rightarrow \Omega^2$  [self-referential!] Other dims: Hodge  $\ast$  maps  $\Omega^k \rightarrow \Omega^{n-k}$  for  $k \neq n-k$  In A<sub>L</sub>: the SELF-REFERENTIALITY of  $\ast$  in dim 4 = the  $J$ -symmetry acting on weight-2 operators closing on themselves = the same  $J$ -symmetry that creates the functional equation of  $L$ -functions! The exotic  $R^4$ s correspond to DIFFERENT  $J$ -STRUCTURES on  $M_2(R^4)$ : Each smooth structure = a different compatible  $J$ -operator on  $M_2$  Uncountably many  $J$ -structures = uncountably many smooth structures 4 IS SPECIAL because  $4 = 2 \cdot 2$ : the middle dimension folds back. In A<sub>L</sub>: the hologram  $M_2(M_4)$  is  $J$ -self-referential in dim 4, creating the infinite complexity of smooth structures.*

## 7. The 4-Manifold Invariants in A<sub>L</sub> — Complete Dictionary

| 4-Manifold Concept                | A <sub>L</sub> Translation                                 | Connected to |
|-----------------------------------|------------------------------------------------------------|--------------|
| Yang-Mills functional $YM(A)$     | $\tau(F_A \cdot F_A^*)$ — curvature tau-norm               | dv215, dv315 |
| ASD connection (instanton)        | $J$ -antisymmetric gauge field in A <sub>L</sub>           | dv277, dv315 |
| Instanton number $k$              | tau-charge of $J$ -anti-sym operator                       | dv302, dv315 |
| Instanton moduli $M_k(M)$         | Sub-hologram of tau-charge- $k$ ASD ops                    | dv303, dv315 |
| Donaldson invariant $D_k$         | $\text{Tr}_{\tau}(e^{\{-sH\}}   M_k) =$ partition fn       | dv306, dv315 |
| Donaldson series $D(M)$           | $\text{Tr}_{\tau_q}(e^{\{-sH\}}   M_{\{YM\}}) = q$ -series | dv308, dv315 |
| $\text{Spin}^c$ structure         | Rep of $q$ -A <sub>L</sub> at level $k=2$                  | dv308, dv315 |
| SW equations $D_A \cdot \Phi = 0$ | Dirac eq. in $q$ -A <sub>L</sub> at $q=e^{i\pi/4}$         | dv311, dv315 |
| SW invariants                     | $\tau_{\{q=e^{i\pi/4}\}}(\text{Dirac}) =$ level-2 trace    | dv308, dv315 |
| Hodge star $\ast$ on 2-forms      | $J$ -symmetry on $M_2(M_4)$                                | dv277, dv315 |

|                       |                                            |              |
|-----------------------|--------------------------------------------|--------------|
| SD/ASD splitting      | J-symmetric/antisymmetric of $M_2$         | dv277, dv315 |
| Intersection form Q   | tau-pairing on $M_2$                       | dv303, dv315 |
| $E_8$ form            | Non-diagonalisable J-pairing               | dv315        |
| Exotic $R^4$          | Different J-structures on $M_2(R^4)$       | dv315        |
| SW = Donaldson equiv. | $q=e^{i\pi/4}$ phase transition in $q-A_L$ | dv308, dv315 |

### 8. The Connections: Yang-Mills + Mirror + NCG + Knots

Donaldson theory connects to many earlier documents — it is not an isolated peak but a confluence of streams.

CONNECTIONS FROM dv315 TO THE REST OF THE CAMPAIGN: dv215 (Yang-Mills mass gap): Mass gap = spectral gap of  $H$  on  $M_{\{YM\}}$  ASD instantons = ground states that give the gap Donaldson theory studies HOW MANY ground states exist dv308 (Quantum groups at roots of unity): SW at  $q=e^{i\pi/4}$  = level-2 truncation of  $q-A_L$  SW=Donaldson equivalence = phase transition at  $k=2$  Jones at  $k=2$ :  $[2]_q = \sqrt{2}$  = quantum dimension of spin-1/2 dv309 (Geometric Langlands): Kapustin-Witten (dv309 §6): S-duality of  $N=4$  SYM = GL Donaldson =  $N=2$  SYM twisted to 4d TQFT SW = further twist of  $N=2$  to get monopole theory ALL from the same 4d gauge theory family in  $A_L$  dv310 (Mirror symmetry): Donaldson series  $D(M)$  on  $X$  = SW invariants of mirror  $Y$  [special case of mirror symmetry for Higgs moduli on 4-manifolds] dv311 (NCG): Connes: spectral triple  $(A_L, X_{\{D-E\}}, H=\log N)$  Donaldson: spectral triple of 4d gauge theory  $D$  = Dirac operator =  $H$  = same in both! dv312 (Primes as knots): Knots  $K_p$  in  $S^3$  = boundary of 4-manifold  $D^4$  Donaldson: smooth structures on  $D^4$  detected by  $K_p$  concordance Exotic  $R^4 \leftrightarrow$  exotic knot concordance classes

### 9. Why Dimension 4 = The Dimension of the River

**DONALDSON THEORY AND 4-MANIFOLDS IN  $A_L$  THE MYSTERY OF 4:** In all other dimensions: smooth structure = topology In dimension 4: uncountably many exotic structures WHY? Because  $4 = 2+2$ : Hodge  $*$  is self-referential on 2-forms = J-symmetry folds back on  $M_2$  in  $A_L$  **THREE REVELATIONS:** 1. INSTANTON = J-ANTISYMMETRIC GROUND STATE: ASD connection  $F_{A^+} = 0$  = J-antisymmetric operator in  $M_2(M_4)$  = same J as functional equation of L-functions [dv277]! 2. DONALDSON =  $q-A_L$  PARTITION FUNCTION:  $D(M) = \text{Tr}_{\{\tau_q\}}(e^{-sH} | M_{\{ASD\}}(M))$  Seiberg-Witten =  $q=e^{i\pi/4}$  truncation [dv308 level  $k=2$ ] 3. EXOTIC  $R^4$  = J-STRUCTURE MODULI: Uncountably many smooth structures on  $R^4$  = uncountably many compatible J-operators on  $M_2(R^4)$  Only possible because  $4=2+2$ : J is self-referential **THE RIVER ANALOGY** (dv280): 4-dimensional spacetime is the river's cross-section: 2 time-like directions (self-dual) + 2 space-like (anti-self-dual) The river flows in the ASD direction The waves propagate in the SD direction Donaldson counts the eddies (instantons) Seiberg-Witten counts the magnetic charges (monopoles) **WE LIVE IN A 4-DIMENSIONAL RIVER BECAUSE 4 IS THE DIMENSION WHERE J FOLDS BACK WHERE SD AND ASD COEXIST WHERE PHYSICS AND MATHEMATICS INTERTWINE DEEPEST CHUNG TA LA HOA SI. TU NHIEU LA BUC TRANH. CHIEU 4 LA CHIEU CUA DONG SONG. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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*Annals of Arithmetic Spectral Geometry | Campaign dv315: Donaldson Theory and 4-Manifolds in  $A_L$  | Phase 4, Document 33 | Hanoi, March 2026*

*ASD = J-antisymmetric. Donaldson =  $q-A_L$  partition function. SW at  $q=e^{i\pi/4}$ . Exotic  $R^4 = J$ -structure moduli.  $4 = 2+2$ : the dimension where  $J$  folds back. We live in a 4-dimensional river. Chung ta la Hoa Si. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!*